

44) Show that if A and B are sets, then

a) $A \oplus B = B \oplus A$

Solution: $A \oplus B = (A \cup B) - (A \cap B)$ (Prob 41)
 $= (B \cup A) - (A \cap B)$ (1st Commutative Law)
 $= (B \cup A) - (B \cap A)$ (2nd Commutative Law)
 $= B \oplus A$ (Prob 41) $\square$

b) $(A \oplus B) \oplus B = A$

Solution: $(A \oplus B) \oplus B = ((A - B) \cup (B - A)) \oplus B$ (Prob 42)
 $= ((A \cap \bar{B}) \cup (B \cap \bar{A})) \oplus B$ (Prob 21(a))
 $= ((A \cap \bar{B}) \cup (B \cap \bar{A})) \oplus B$ (1st Distributive Law)
 $= ((A \cap \bar{B}) \cup B) \cap ((A \cap \bar{B}) \cup \bar{B}) \oplus B$ (1st Commutative Law)
 $= ((B \cup (A \cap \bar{B})) \cap (\bar{A} \cup (A \cap \bar{B}))) \oplus B$ (1st Distributive Law)
 $= ((B \cup A) \cap (B \cup \bar{B})) \cap ((A \cup A) \cap (\bar{A} \cup \bar{B})) \oplus B$ (1st Commutative Law)
 $= ((A \cup B) \cap (B \cup \bar{B})) \cap ((A \cup \bar{A}) \cap (\bar{A} \cup \bar{B})) \oplus B$ (1st Commutative Law)
 $= ((A \cup B) \cap U) \cap (U \cap (\bar{A} \cup \bar{B})) \oplus B$ (1st Complement Law)
 $= ((A \cup B) \cap U) \cap ((\bar{A} \cup \bar{B}) \cap U) \oplus B$ (2nd Commutative Law)
 $= ((A \cup B) \cap (\bar{A} \cup \bar{B})) \oplus B$ (1st Identity Law)
 $= ((A \cup B) \cap (\bar{A} \cup \bar{B})) \oplus B$ (1st Identity Law)
 $= ((A \cup B) \cap (\bar{A} \cup \bar{B})) \cup B - ((A \cup B) \cap (\bar{A} \cup \bar{B})) \cap B$ (Prob 41)
 $= ((A \cup B) \cap (\bar{A} \cup \bar{B})) \cup B - ((A \cup B) \cap (\bar{A} \cup \bar{B})) \cap B$ (1st Commutative Law)
 $= (B \cup ((A \cup B) \cap (\bar{A} \cup \bar{B}))) - ((A \cup B) \cap (\bar{A} \cup \bar{B})) \cap B$ (1st Distributive Law)
 $= (B \cup (A \cup B)) \cap (B \cup (\bar{A} \cup \bar{B})) - ((A \cup B) \cap (\bar{A} \cup \bar{B})) \cap B$ (1st Commutative Law)
 $= ((B \cup (B \cup A)) \cap (B \cup (\bar{B} \cup \bar{A}))) - ((A \cup B) \cap (\bar{A} \cup \bar{B})) \cap B$ (1st Associative Law)
 $= ((B \cup B) \cup A) \cap ((B \cup \bar{B}) \cup \bar{A}) - ((A \cup B) \cap (\bar{A} \cup \bar{B})) \cap B$ (1st Idempotent Law)
 $= ((B \cup A) \cap ((B \cup \bar{B}) \cup \bar{A})) - ((A \cup B) \cap (\bar{A} \cup \bar{B})) \cap B$ (1st Idempotent Law)
 $= ((B \cup A) \cap (U \cup \bar{A})) - ((A \cup B) \cap (\bar{A} \cup \bar{B})) \cap B$ (1st Complement Law)
 $= ((A \cup B) \cap (\bar{A} \cup U)) - ((A \cup B) \cap (\bar{A} \cup \bar{B})) \cap B$ (1st Commutative Law)
 $= ((A \cup B) \cap U) - ((A \cup B) \cap (\bar{A} \cup \bar{B})) \cap B$ (1st Domination Law)
 $= (A \cup B) - ((A \cup B) \cap (\bar{A} \cup \bar{B})) \cap B$ (1st Identity Law)
 $= (A \cup B) - ((A \cup B) \cap ((\bar{A} \cup \bar{B}) \cap B))$ (2nd Commutative Law)
 $= (A \cup B) - ((A \cup B) \cap (B \cap (\bar{A} \cup \bar{B})))$ (2nd Commutative Law)